

ENGINEERING SPECIFICATIONS

Document Name: KDC Double Circuit Modbus List		
Document No: ESP-24-25-10	Revision: R01	Date:26-09-2024
Prepared By: PJP	Checked By: AS	Approved By: NMR

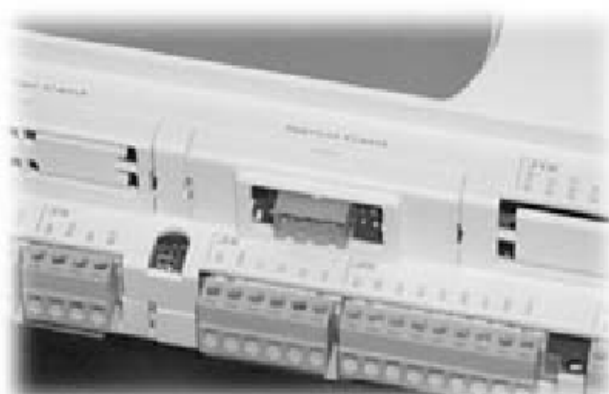
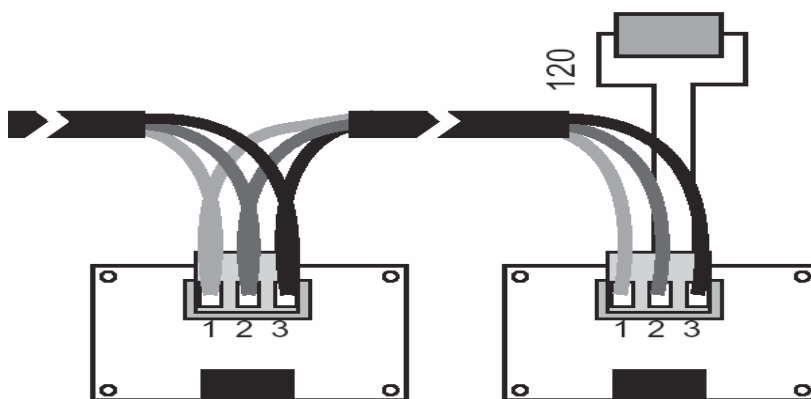
KDC Double Circuit				
Descriptions	UOM	Type	Read/Write	Modbus Add(Holding Register)
Evaporator Water In Temperature	°C	ANALOG	Read	40005
Evaporator Water Out Temperature	°C	ANALOG	Read	40006
Condenser IN Temperature/Ambient	°C	ANALOG	Read	40014
Condenser Out Temperature	°C	ANALOG	Read	40015
LWT Setpoint	°C	ANALOG	Read/Write	40051
Suction Pressure	Bar	ANALOG	Read	40002
Evaporator Approach	-	INTEGER	Read	40025
Compressor - 1				
Discharge Temperature	°C	ANALOG	Read	40009
Liquid Line Temperature	°C	ANALOG	Read	40013
Discharge Pressure	Bar	ANALOG	Read	40003
Motor Current	Amp	ANALOG	Read	40007
Suction Superheat	°C	ANALOG	Read	40017
Suction Temperature	°C	ANALOG	Read	40012
Percentage RLA	%	ANALOG	Read	40031
Condenser Approach	-	INTEGER	Read	45028
Run Hours- High*	Hrs.	INTEGER	Read	45105
Run Hours- Low*	Hrs.	INTEGER	Read	45104
Number of Starts H*	-	INTEGER	Read	45111
Number of Starts L*	-	INTEGER	Read	45114
Compressor - 2				
Discharge Temperature	°C	ANALOG	Read	40037
Liquid Line Temperature	°C	ANALOG	Read	40041
Discharge Pressure	Bar	ANALOG	Read	40033
Motor Current	Amp	ANALOG	Read	40035
Suction Superheat	°C	ANALOG	Read	40042
Suction Temperature	°C	ANALOG	Read	40039
Percentage RLA	%	ANALOG	Read	40141
Condenser Approach	-	INTEGER	Read	45051
Run Hours- High*	Hrs.	INTEGER	Read	45116
Run Hours- Low*	Hrs.	INTEGER	Read	45115
Number of Starts H*	-	INTEGER	Read	45120
Number of Starts L*	-	INTEGER	Read	45123

ENGINEERING SPECIFICATIONS

Document Name: KDC Double Circuit Modbus List		
Document No: ESP-24-25-10	Revision: R01	Date:26-09-2024
Prepared By: PJP	Checked By: AS	Approved By: NMR

Parity	None
Baud rate	19200
Protocol	Modbus/Modbus Extended
Start bit	1
Stop bit	1
No of bit	8
Communication Interface	RS-485
Interface Connection	2 Wire

The connection with the RS485 network is carried out using the plug-in terminal connector on the card. Pin-wiring of the connector is stamped on the card. If the card is placed in the last position of the supervision serial line, pins 2 and 3, you must connect a 120 Ω - 1/4 W end line resistor, as shown.



pin	significato <i>meaning</i>
1	GND
2	RX+/TX+
3	RX-/TX-

*Total Run hours/No of starts will be calculated as below *Total run hrs = Run Hours- High*x 1000 + Run Hour Low
Total no of starts = Number of starts H x 1000 + Number of starts L